PROJECT PRESENTATION

Polynomial Regression for School Attendance

Predicting students present from students enrolled

Python • Pandas • Scikit-learn



Project Overview

What does this project try to do?

The problem

Understand the relationship between student enrollment and daily attendance.

The input

Enrolled — the number of students enrolled.

The target

Present — the number of students who attended.

Main idea: fit a curve to historical data and estimate attendance.

Objectives

Key goals of the analysis

- Load and inspect the school attendance CSV dataset.
- Use Enrolled as the input feature (X).
- Use Present as the target (y).
- Train a degree-2 Polynomial Regression model.
- Evaluate predictions using MSE and R^2.
- Visualize observed data and the fitted curve.

Dataset & Tools

Data source and main Python libraries

Dataset file

2018-2019_Daily_Attendance_20240429.csv

Pandas + NumPy

Load data, work with columns and arrays, and generate values.

Scikit-learn

Split data, create polynomial features, train and evaluate the model.

Matplotlib + Seaborn

Matplotlib creates the chart; Seaborn is imported but not used in the notebook.

Required columns: Enrolled and Present

Project Workflow

From raw data to model evaluation

1 Load CSV

Read data with Pandas

2 Select X and y

Enrolled → Present

3 Split data

80% train / 20% test

4 Create features

Polynomial degree = 2

5 Train model

Linear Regression

6 Evaluate + plot

MSE, R^2 and chart

What Is Polynomial Regression?

A curve can model relationships that are not perfectly straight

Linear model

A straight line captures a constant rate of change.

Polynomial model

Adds powers such as x^2 so the fitted relationship can bend.

Degree-2 model form

$$\hat{y} = b_0 + b_1x + b_2x^2$$

x = Enrolled students • $\hat{y}$ = predicted Present students

Training and Testing

Why split the dataset?



Training set

Used to fit the model.

Testing set

Checks predictions on unseen rows.

random_state = 42 makes the split repeatable.

Evaluation Metrics

How do we judge the model?

Mean Squared Error (MSE)

Measures average squared difference between actual and predicted attendance. Lower is generally better.

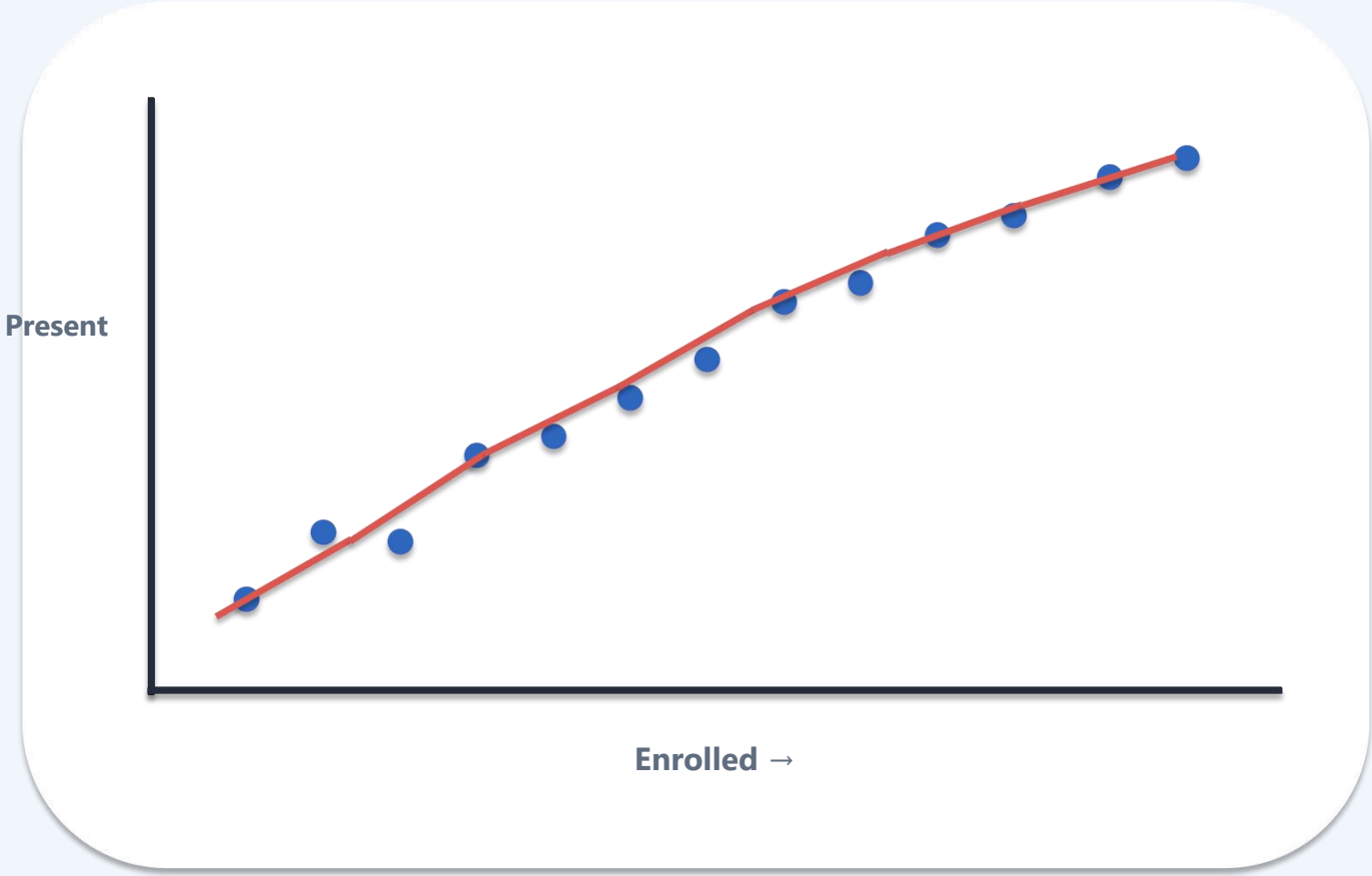
R^2 Score

Shows how much variation in the target is explained by the model. Closer to 1 often indicates a better fit.

The notebook prints the actual scores when executed. No numeric results are assumed here.

Visualization

What the plot represents



Conceptual illustration — not measured output

Blue points

Sampled actual observations.

Red curve

Model's predicted relationship.

Code Note & Improvement

A safer way to sample test rows for plotting

Potential issue

Selecting 1,000 rows without replacement can fail if the test set has fewer than 1,000 rows.

```
sample_size = min(1000, len(X_test))
indices = np.random.choice(len(X_test), size=sample_size, replace=False)
```

Also check missing values and confirm both columns are numeric before training.

Conclusion

What this project demonstrates

- An introductory end-to-end machine-learning workflow.
- Polynomial features allow a regression model to fit a curved relationship.
- MSE and R^2 help evaluate predictions on held-out data.
- Usefulness depends on data quality and measured scores.
- A relationship alone does not explain why attendance changes.

THANK YOU

Questions?

Polynomial Regression for School Attendance